

Calc 2 – Sequences

Worksheet A

1. Determine $\lim_{n \rightarrow \infty} \frac{2n^2 + 3n}{n^2 + 1}$.
2. Determine whether $\{a_n\}$ with $a_n = (-1)^n$ converges.
3. Use the squeeze theorem to find $\lim_{n \rightarrow \infty} \frac{\cos n}{n}$.
4. Find $\lim_{n \rightarrow \infty} \sqrt[n]{n}$ (use $\ln$).
5. Show that $a_n = \frac{n}{n+1}$ is increasing and bounded above, then find its limit.

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Worksheet B

1. Determine $\lim_{n \rightarrow \infty} \frac{3^n}{n!}$.
2. Determine whether $\{a_n\}$ with $a_n = \sin(n\pi/2)$ converges.
3. Use the squeeze theorem to find $\lim_{n \rightarrow \infty} \frac{n!}{n^n}$.
4. Find $\lim_{n \rightarrow \infty} \left(1 + \frac{3}{n}\right)^n$.
5. Show $a_n = \sqrt{2 + a_{n-1}}$, $a_1 = \sqrt{2}$, is bounded above by 2 and increasing, then find $\lim a_n$.

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Answer Key (Worksheets A & B)

Worksheet A.

1. Divide by n^2 : $\frac{2+3/n}{1+1/n^2} \rightarrow \boxed{2}$.
2. a_n alternates $-1, 1, -1, \dots$, so it does not approach a single value. **Diverges.**
3. $-1/n \leq \cos n/n \leq 1/n$, both bounds $\rightarrow 0$, so limit = $\boxed{0}$.
4. $\ln(\sqrt[n]{n}) = \frac{\ln n}{n} \rightarrow 0$, so $\sqrt[n]{n} \rightarrow e^0 = \boxed{1}$.
5. $a_{n+1} - a_n = \frac{n+1}{n+2} - \frac{n}{n+1} = \frac{1}{(n+1)(n+2)} > 0$, increasing. Bounded above by 1. Limit = $\boxed{1}$.

Worksheet B.

1. Ratio: $\frac{a_{n+1}}{a_n} = \frac{3}{n+1} \rightarrow 0 < 1$, so $a_n \rightarrow \boxed{0}$ (standard fact: factorials beat exponentials).
2. Values cycle $1, 0, -1, 0, 1, \dots$. **Diverges.**
3. $0 < \frac{n!}{n^n} = \frac{1}{n} \cdot \frac{2}{n} \cdots \frac{n}{n} \leq \frac{1}{n} \cdot 1 \cdots 1 = \frac{1}{n} \rightarrow 0$. So limit = $\boxed{0}$.
4. Standard limit: $\boxed{e^3}$.
5. Suppose $a_n < 2$. Then $a_{n+1} = \sqrt{2+a_n} < \sqrt{4} = 2$. By induction, bounded by 2. Also $a_{n+1}^2 - a_n = 2 + a_n - a_n^2 = -(a_n - 2)(a_n + 1) > 0$ for $a_n < 2$, so increasing. Limit L satisfies $L = \sqrt{2+L} \Rightarrow L^2 - L - 2 = 0 \Rightarrow \boxed{L=2}$.